

科技信息检索与利用调研报告

——以“人形机器人运动控制”为例

一、选题简介

人形机器人（Humanoid Robot）是指具备类人形态结构并能够模拟人类运动行为的智能机器人系统，被誉为机器人领域的“皇冠上的明珠”。近年来，随着人工智能、先进传感技术与制造工艺的快速发展，人形机器人研究进入新一轮爆发期。宇树 G1、波士顿动力 Atlas、特斯拉 Optimus 等产品相继亮相，Figure AI、Agility Robotics 等初创企业获得大量资本关注，人形机器人正加速从实验室走向产业化应用。

运动控制作为人形机器人的核心技术，直接决定其行走稳定性、运动灵活性及对复杂环境的适应能力。相较于轮式或多足机器人，人形机器人采用双足直立行走方式，具有支撑面积小、重心高、自由度多等显著特征，使其在动态平衡、步态协调与快速恢复方面面临更为严峻的控制挑战。如何实现稳定高效的步态规划、精准快速的平衡调节、全身多自由度的协调运动，已成为当前研究的关键问题。从技术发展路径看，人形机器人运动控制经历了从基于模型的传统控制方法向基于学习的智能控制方法的演进，展现出了强大的应用潜力。

基于此，本次文献调研旨在系统梳理人形机器人运动控制领域的研究现状与发展趋势，通过综合分析，全面总结相关文献的特点与差异，比较国内外代表性研究成果，识别主要研究机构、学者以及高水平期刊和重要学术会议的研究重点。在此基础上，进一步聚焦强化学习在运动控制中的应用前沿方向，提炼当前研究热点与未来发展趋势，为后续深入研究和选题提供参考依据。

二、调研工具与其组合选用逻辑简介

本调研采用多数据库、多工具协同的组合式检索策略，以兼顾覆盖广度、学术权威性、前沿时效性与技术应用深度。

首先，在检索关键词确定阶段选用 Google Scholar 作为起点，其覆盖范围最广、跨学科特性突出，能够在调研初期快速浏览高被引文献与代表性工作，从而提炼领域内的核心概念、常用术语与关键研究方向，为后续系统检索奠定统一的关键词体系。

在此基础上，针对核心研究人员、研究机构与学术力量的识别，引入 Web of Science 这一权威引文索引数据库，利用其成熟的作者、机构与引文分析功能，通过核心关键词的系统检索与结果分析，客观识别在该研究领域内具有较高影响力的学者、课题组及研究机构，确保研究对象的学术代表性与可靠性。

随后，在核心前沿研究检索阶段，重点结合 IEEE Xplore 与 arXiv 两个平台，其中

IEEE Xplore 系统收录机器人领域的顶级期刊与国际会议论文，是追踪核心学者正式发表成果与技术路线演进的关键来源，而 arXiv 则以其高度的时效性弥补传统期刊发表周期较长的不足，有助于把握最新研究动态和尚未正式发表的前沿工作。

对于人形机器人新兴产业，企业在相关技术方向上的布局与发展十分关键，因此进一步引入 Google Patents，对 Tesla、Figure AI、宇树科技等代表性企业进行申请人维度的专利检索，从产业视角分析其在人形机器人运动控制等关键技术上的研发重点与演进方向。

与此同时，考虑到上述检索策略具有方法论层面的普遍适用性，但对"人形机器人运动控制"这一特定课题而言，可能难以同时覆盖研究成果的时效性和有效性。因此，有必要通过阅读领域综述文献、参考 GitHub 上动态更新的文献仓库等方式进行针对性补充。

最后，在深度拓展阶段选用 Connected Papers，对阅读过程中具有代表性的核心文献进行可视化关联分析，通过引用网络与语义相似性挖掘相邻研究领域的相关工作，从而在既定研究主线的基础上实现系统性的纵向深化与横向拓展。

整体而言，调研的检索流程形成了由“宽泛探索—权威筛选—前沿跟踪—产业补充—深度拓展”逐层递进的逻辑结构，能够较为全面、系统地支撑人形机器人运动控制领域的文献调研与研究现状分析。

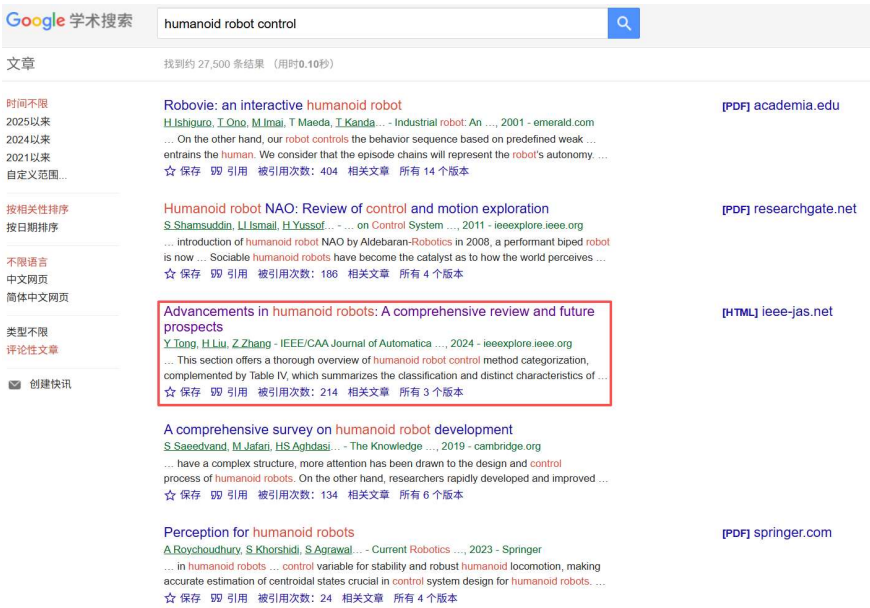
检索阶段	数据库/工具	选择依据	选择目标
检索关键词确定	Google Scholar	覆盖范围最广的学术搜索引擎，适合调研初期快速探索领域、收集核心关键词和术语体系，使用门槛低、效率高	初步检索宽泛词汇，浏览高被引论文，收集领域核心关键词
核心研究人员、组织、企业检索	Web of Science	全球最权威的引文索引数据库，机构分析和作者分析功能强大，可系统识别领域内的顶尖组织和核心学者	使用核心关键词进行系统检索，利用"分析结果"功能识别影响力高的学者
核心前沿研究检索	IEEE Xplore arXiv	IEEE Xplore 完整收录机器人领域顶级会议和核心期刊，是追踪课题组学术成果的最佳平台； arXiv 是时效性最强的预印本平台，是追踪前沿研究动态的必备资源	针对识别出的核心学者，进行作者检索，获取其在 IEEE 期刊、会议以及预印本上的全部论文
企业专利调研	Google Patents	免费开放的全球专利检索平台，覆盖 100 多个国家专利数据，申请人检索功能便捷，是调研企业技术布局的理想工具	以 Tesla、Figure AI、宇树科技等企业名称为申请人，检索其运动控制相关专利

零散研究 补充	综述文献 Github 仓库	综述文献和 Github 文 献仓库由领域专家撰写维 护，可弥补关键词检索的 盲区	从综述文献中选取感 兴趣领域的内容精读，拓 展阅读关键词搜索未查找 到的文献
深度拓展	Connected Papers	创新的可视化文献分 析工具，通过语义分析和 引用网络发现相似领域论 文，适合对感兴趣的核心 文献进行深度拓展，发现 相邻领域的相关工作	选取阅读过程中特别 感兴趣的核心论文，生成 文献关联图谱，发现相似 领域的相关研究和技术路 线

三、调研过程

1. 检索关键词确定

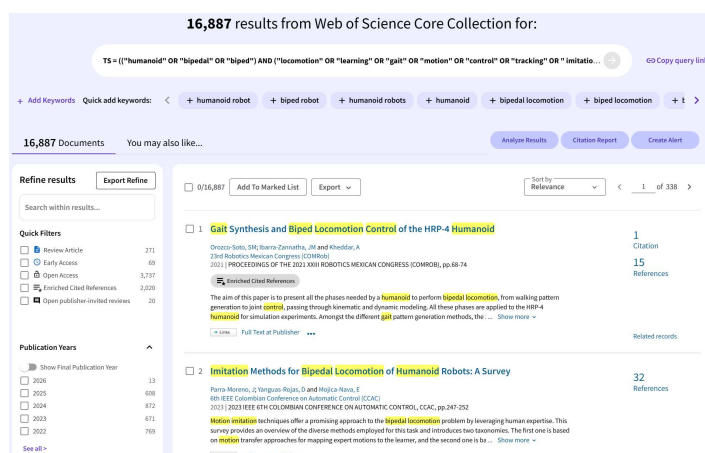
在 Google scholar 粗略搜索 humanoid robot control 关键词，结合时效性和被引次数，选取一篇合适论文，进行核心关键词提取。如下图所示，选取 Advancements in Humanoid Robots: A Comprehensive Review and Future Prospects 综述性文章为切入点。



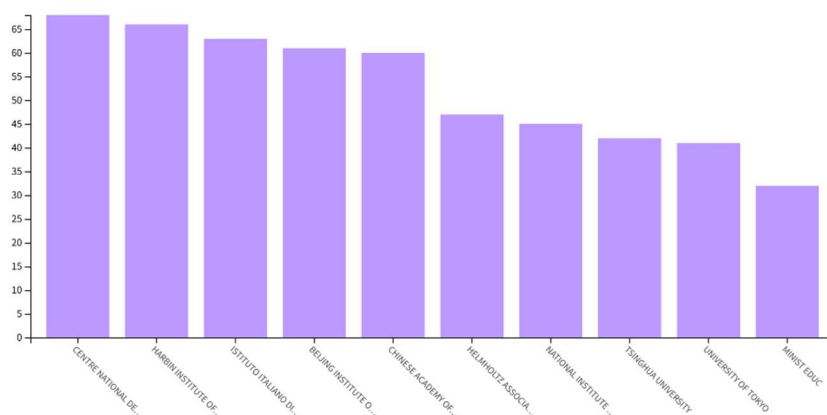
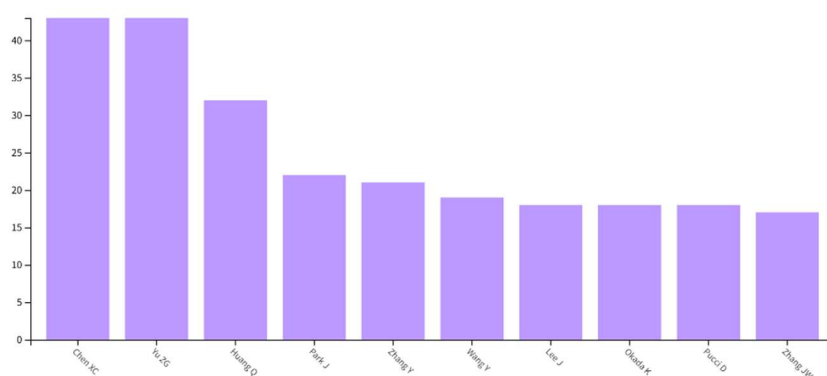
结合豆包 AI 工具，在文章中提取到详细关键词，如 locomotion、bipedal、whole-body control、locomotion、learning、imitation、tracking 等。

2. 核心研究人员、组织检索

根据关键词，构建 WOS 检索式 TS = (("humanoid" OR "bipedal" OR "biped") AND ("locomotion" OR "learning" OR "gait" OR "motion" OR "control" OR "tracking" OR "imitation"))进行检索，得到结果如下。



由于新兴领域方法迭代迅速，勾选近 3 年研究内容，分析搜索结果，得到分析结果如下。



综合文章质量和数量，选择继续跟踪 Chen XC 以及 Tsinghua University 的相关工作。

3. 核心前沿研究检索

在 IEEE Xplore 和 arXiv 中分别以作者关键词搜索 Chen Xuechao (Chen XC 全名)，得到搜索结果如下。

Showing 1-25 of 40 results for "First Name": Xuechao × "Last Name": Chen ×
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☐ A Single-arm Mode for Operating Steering Wheel in Robot Driving

Xin Zhu; Xuechao Chen; Kehong Chen; Yue Dong; Zhaoyang Cai; Zhuo Chen
2024 IEEE International Conference on Mechatronics and Automation (ICMA)
Year: 2024 | Conference Paper | Publisher: IEEE

▼ AbstractHTMLPDFCC

☐ TGH-Planner: Topology-Guided Hierarchical Planner for Nonholonomic Robots in 2-D Unknown and Complex Environments

Xuechao Chen; Hengbo Qi; Zhangguo Yu; Zeyu Zhang; Ruiwen Yang; Yongliang Shi; Fei Meng; Qiang Huang
IEEE/ASME Transactions on Mechatronics
Year: 2025 | Early Access Article | Publisher: IEEE

▼ AbstractPDFCC

☐ Online Running Pattern Generation for Humanoid Robot With Direct Collocation of Reference-Tracking Dynamics

Huanzhong Chen; Xuechao Chen; Chencheng Dong; Zhangguo Yu; Qiang Huang
IEEE/ASME Transactions on Mechatronics
Year: 2024 | Volume: 29, Issue: 3 | Journal Article | Publisher: IEEE
Cited by: Papers (4)

▼ AbstractHTMLPDFCC

☐ Versatile Bipedal Locomotion and Walking-Running Transition: Coordinating Supervised Learning and Nonlinear Optimization

Huanzhong Chen; Gao Huang; Xuechao Chen; Zhangguo Yu; Chencheng Dong; Qingqing Li; Qiang Huang
IEEE Transactions on Automation Science and Engineering
Year: 2025 | Volume: 22 | Journal Article | Publisher: IEEE

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Chen, Xuechao

Author(s)

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1. arXiv:2506.12314 [pdf, ps, other] cs.RD eess.SY

Explosive Output to Enhance Jumping Ability: A Variable Reduction Ratio Design Paradigm for Humanoid Robots Knee Joint

Authors: Xiaoshuai Ma, Haoxiang Qi, Qingqing Li, Haochen Xu, Xuechao Chen, Junyao Gao, Zhangguo Yu, Qiang Huang

Submitted 13 June, 2025; originally announced June 2025.

2. arXiv:2502.00218 [pdf] cond-mat.mes-hall physics.app-ph physics.optics quant-ph doi 10.1364/OPTICA.569882

Waveguide Excitation and Spin Pumping of Chirally Coupled Quantum Dots

Authors: Savvas Germanis, Xuchao Chen, René Dost, Dominic J. Hallett, Edmund Clarke, Pallavi K. Patil, Maurice S. Skolnick, Luke R. Wilson, Hamidreza Siampour, A. Mark Fox

Submitted 31 January, 2025; originally announced February 2025.

Comments: 11 pages, 4 figures

Journal ref: Optica 12, 1689-1696 (2025)

3. arXiv:2501.12594 [pdf, other] cs.RD doi 10.1109/TASE.2025.3557162

A 3-Step Optimization Framework with Hybrid Models for a Humanoid Robot's Jump Motion

Authors: Haoxiang Qi, Zhangguo Yu, Xuechao Chen, Yaliang Liu, Chuanku Yi, Chencheng Dong, Fei Meng, Qiang Huang

Submitted 21 January, 2025; originally announced January 2025.

4. arXiv:2408.10007 [pdf, other] cs.CV

P3P: Pseudo-3D Pre-training for Scaling 3D Voxel-based Masked Autoencoders

Authors: Xuechao Chen, Ying Chen, Jialin Li, Qiang Nie, Hanqiu Deng, Yong Liu, Qixing Huang, Yang Li

Submitted 21 May, 2025; v1 submitted 19 August, 2024; originally announced August 2024.

Comments: Under review. Pre-print

5. arXiv:2404.18047 [pdf, other] cs.RD

在 IEEE Xplore 和 arXiv 检索过程中会遇到作者重名的问题，需要手动剔除。该问题可以通过 Web of Science 的详情页跳转替换，但是更新时效性不如 arXiv，需要进一步分析辨别。

同时，由于 Tsinghua University 的搜索范围过于宽泛，也直接使用 WoS 跳转到详情页，可以得到检索结果如下。

TS = ("humanoid" OR "bipedal" OR "biped") AND ("locomotion" OR "learning" OR "gait" OR "motion" OR "control" OR "tracking" OR "imitatio...

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Document Types

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- ☐ Early Access
- ☐ Review Article

Researcher Profiles

- ☐ Show Researcher Profiles
- ☐ Ming Zhao
- ☐ Houde Liu
- ☐ Sun, Tianming
- ☐ Wong, S.C.

1 Fast Terrain-Adaptive Motion of Humanoid Robots Based on Model Reference One-Step-Ahead Predictive Control

Wang, Qi, Li, Q and Zhao, MC
Nov 2023 | IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY • 31(6), pp.2819-2834

6 Citations
57 References

2 Humanoid Parkour Learning

Zhuang, Zhi, Yan, SZ and Zhao, H
8th Conference on Robot Learning, CORL Annual 2024 | CONFERENCE ON ROBOT LEARNING 270

67 References

3 A Joint-Space Humanoid Robot Torso Motion Planning Method Based on Spiking Neural Networks

Ju, Fz, Jin, HZ (-); Zhao, J
Sep 2023 [Early Access] | IEEE-ASME TRANSACTIONS ON MECHATRONICS •

36 References

4. 企业专利调研

对于人形机器人相关产业，由于其前沿产学研结合的特殊性，企业相关研究十分具有代表性，同时企业核心技术通常不以论文形式进行发表，因此在企业层面上，专利方面的检索可以有效替代学术方面检索。

根据豆包 AI 工具推荐以及行业经验筛选，选取了宇树科技和北京人形机器人创新中心有限公司为例，搜索其专利如下。

Google Patents 宇树科技

Assignee: 宇树科技;

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Leg power system structure of electric drive quadruped robot

WO US CN DE • CN110690422B • 王兴兴 • 杭州宇树科技有限公司

Priority 2017-03-10 • Filed 2017-03-10 • Granted 2022-12-23 • Published 2022-12-23

The invention relates to a leg structure of a quadruped robot, in particular to a leg power system structure of an electrically driven quadruped robot. The structure of the robot legs in the prior art is not highly integrated, and the resistance to falling is not strong. The present invention is ...

A highly integrated and high-performance robot joint unit

WO EP US CN JP • CN10591045B • 王兴兴 • 杭州宇树科技有限公司

Priority 2018-12-20 • Filed 2018-12-20 • Granted 2025-01-28 • Published 2025-01-28

The present invention discloses a highly integrated and high-performance robot joint unit, belonging to the field of robot equipment technology. When the reducer of the existing robot joint is in an overload working state, it will cause damage to the reducer gear and other joint parts, and the ...

Integrated joint module for robot and walking robot applying it

WO EP US CN JP • JP7245588B2 • 王兴兴 • 杭州宇树科技有限公司

Priority 2019-03-07 • Filed 2020-02-24 • Granted 2023-03-24 • Published 2023-03-24

comprising a first motor and reducer assembly (3), a second motor assembly (4), a second reducer assembly (40) and a first output link (1); The first motor/reducer assembly (3) has the first output link (1) attached to its output shaft end side, and the second motor is attached to the other side ...

A kind of linear drive apparatus of the high power mass density for simplifying ...

WO CN DE • CN20631359B • 王兴兴 • 杭州宇树科技有限公司

Priority 2017-01-12 • Filed 2017-01-12 • Granted 2021-07-07 • Published 2021-07-07

The utility model discloses a linear drive device with a simplified structure and high power mass density, which belongs to the technical field of linear drive equipment for robot joints. The structure of the screw nut driving unit in the prior art is relatively complicated, the number of parts is ...

A kind of electric drive rotary motive power unit and a kind of quadruped robot ...

CN • CN20884568B • 王兴兴 • 杭州宇树科技有限公司

Priority 2018-08-07 • Filed 2018-08-07 • Granted 2019-05-10 • Published 2019-05-10

The utility model relates to the technical field of robot equipment, and discloses an electric drive rotary power unit a robot joint and a quadruped robot applying the same. The utility model comprises a motor base, a motor unit and a reducer unit, wherein the motor unit and the reducer unit ...

A height map construction method, system and storage medium for robots

WO CN • CN11633851B • 孟令波 • 杭州宇树科技有限公司

Priority 2023-02-24 • Filed 2023-02-24 • Granted 2024-02-23 • Published 2024-02-23

The invention discloses a height map construction method, system and storage medium for robots, and belongs to the technical field of height map construction. The present invention provides a method for constructing a height map for

Google Patents 北京人形机器人创新中心有限公司

Assignee: 北京人形机器人创新中心有限公司;

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A multi-robot collaborative task dynamic scheduling method and related ...

CN • CN115647314A • 任涛 • 北京人形机器人创新中心有限公司

Priority 2025-02-18 • Filed 2025-02-18 • Published 2025-03-18

The present application provides a multi-robot collaborative task dynamic scheduling method and related equipment, which are applied to the field of robot technology, including: collecting task scheduling scene information; using the task scheduling scene information to build a multi-level graph ...

Robot motion control model training method and robot motion control method

CN • CN118466232A • 郭宜弘 • 北京人形机器人创新中心有限公司

Priority 2024-07-15 • Filed 2024-07-15 • Published 2024-08-09

1. A method for training a robot motion control model, comprising: based on the state of the training robot at the current moment, a training control instruction of a preset frame number in the future and a training hidden variable are obtained by utilizing a motion control model to be trained;

Integrated joint

CN • CN118456491B • 祖汪明 • 北京人形机器人创新中心有限公司

Priority 2024-07-08 • Filed 2024-07-08 • Granted 2025-05-02 • Published 2025-05-02

The present application provides an integrated joint, which relates to the field of robot technology. The integrated joint includes a motor, a cycloidal pinwheel reducer, a feedback shaft, a first encoder, a second encoder and a driver, wherein: the driver is electrically connected to the motor to ...

Leg mechanism and humanoid robot

CN • CN222005236U • 肖健 • 北京人形机器人创新中心有限公司

Priority 2025-08-13 • Filed 2025-04-17 • Published 2025-09-13

The application provides a training method and equipment of a scene perception model, a robot control method and a robot, and relates to the technical field of machine learning. The method comprises the steps of obtaining Gaussian geometric features and first semantic features of a plurality of ...

Scene perception model training method, device, robot control method and robot

CN • CN120635678A • 张强 • 北京人形机器人创新中心有限公司

Priority 2025-08-13 • Filed 2025-04-17 • Published 2025-09-12

The application provides a training method and equipment of a scene perception model, a robot control method and a robot, and relates to the technical field of machine learning. The method comprises the steps of obtaining Gaussian geometric features and first semantic features of a plurality of ...

Humanoid robot control method, system and humanoid robot

CN • CN113974029A • 张强 • 北京人形机器人创新中心有限公司

Priority 2025-08-13 • Filed 2025-04-17 • Published 2025-09-12

The application provides a humanoid robot control method, a humanoid robot control system and a humanoid robot, wherein the method comprises the steps of obtaining a task instruction and a visual field image, inputting the visual field

5. 零散研究补充

对于“人形机器人运动控制”课题，在以上关键词搜索后，可以得到题为 Humanoid Locomotion and Manipulation: Current Progress and Challenges in Control, Planning, and Learning 的综述文献，其内容恰好包含所选课题，论文结构如下图所示。

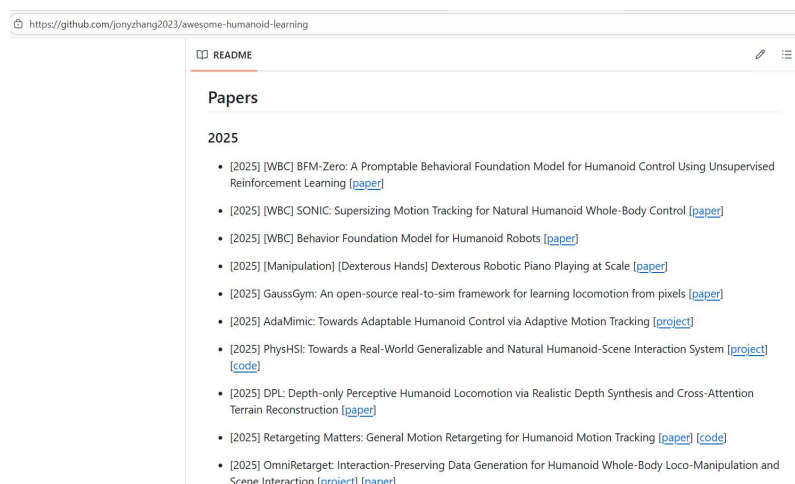


截取综述内的部分内容如下，可以从其简短介绍中提取关键信息并拓展阅读零散的相关文献。但是此方法不具有普遍性，且时效性较差。

3) *Approaches to Learning from Human Data:* Examples of learned human-like motions include walking using Generative Adversarial Imitation Learning (GAIL) [303, 307] and Adversarial Motion Prior (AMP) [304, 305, 326], which is an extension of the GAIL. Recently, RL-based motion imitation using motion capture data has successfully transferred to humanoid robots [267, 268, 324, 327]. However, most of these works achieve only relatively conservative locomotion behaviors. Highly agile behaviors, as shown in DeepMimic [328], still exist only in simulators, and similar capabilities have yet to be replicated with real robots.

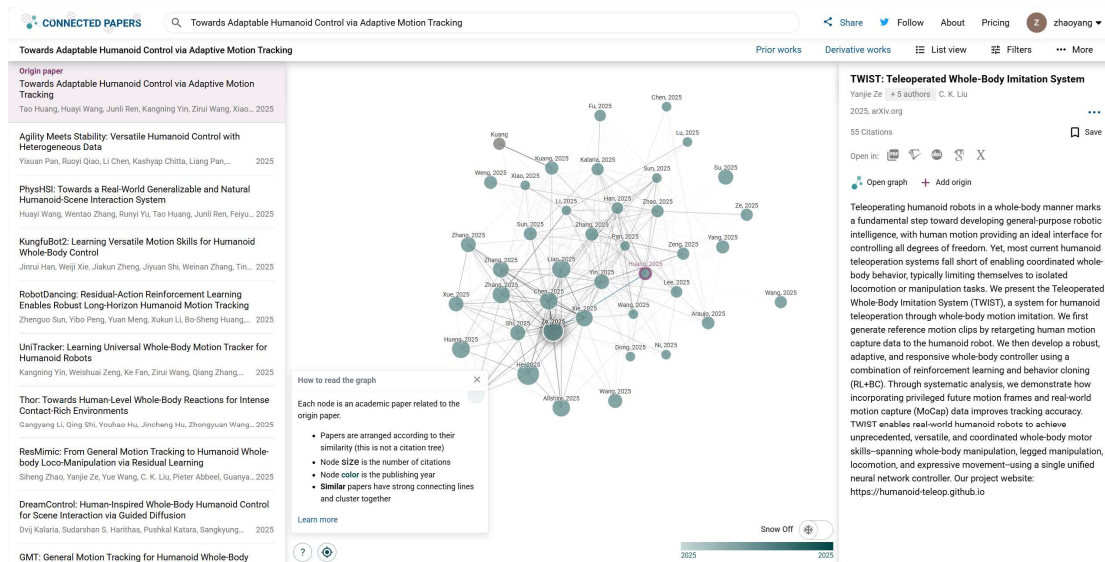
2) *Learning to Track Trajectory Reference:* Combining model-based methods with learning-based methods is extensively explored. Numerous works on quadruped robots [335, 336] use MPC to generate reference motions and use them as imitation rewards for RL-based motion imitation. However, calculating MPC online can prolong the training time and occasionally fail to find a feasible solution. Online RL that tracks *offline-generated* trajectories avoids this problem and has achieved effective RL policies for versatile bipedal locomotion [33, 332, 338]. Supervised learning is also used to mimic TO-generated offline motions. For example, [339] allows humanoid robots to achieve brachiation on monkey bars. In addition to enhancing learning methods with model-informed trajectories, a learned policy can also suggest effective warm starts or hyperparameters for model-based methods, thus significantly reducing the iterations required for convergence to an optimal trajectory.

此外，在 Github 中也有名如“awesome-*”，由具有开源精神的同行专家自发维护的综述文献仓库。就人形机器人领域而言，仓库 [jonyzhang2023/awesome-humanoid-learning](https://github.com/jonyzhang2023/awesome-humanoid-learning): [Humanoid Robots Resources](#) 不断动态更新相关文献，既保证了时效性，也在同行专家挑选下保证了文献重要性，可以补充日常零散阅读的文献。



6. 深度拓展

对于上述粗略筛选阅读后感兴趣的文献，利用 Connected Papers 工具，深度拓展其相似相关的论文，进行进一步调研，了解相关方法的跨领域交叉情况。



上图是 Connected Papers 工具以 Towards Adaptable Humanoid Control via Adaptive Motion Tracking 为例的链接图，分析出了其相似论文的情况，可以进一步阅读摘要与研究。

四、总结与思考

通过本次以“人形机器人运动控制”为主题的文献调研，较为系统地把握了该领域近年的研究脉络与发展趋势。从检索工具与方法的应用效果来看，多数据库协同使用显著提升了调研的系统性与可靠性。Google Scholar 在调研初期对于快速建立领域认知、提炼关键词体系具有不可替代的优势；Web of Science 在核心学者与机构识别、引文分析方面体现出权威性和结构化优势；IEEE Xplore 和 arXiv 的组合有效兼顾了研究成果的学术规范性与前沿时效性；专利数据库的引入则从产业视角补充了论文检索的不足。

然而，各类工具也存在一定局限性，例如关键词检索容易遗漏跨领域或表述差异较大的研究工作，作者重名与机构名称不统一会增加人工筛选成本，综述文献与 GitHub 仓库虽然信息密集，但在系统性和客观性上依赖维护者的主观判断。在此基础上，后续调研仍有进一步优化空间。一方面，可以结合文献计量学方法，引入可视化分析工具，对研究热点演化、方法聚类 and 引用网络进行更定量化的分析；另一方面，可进一步加强对更多交叉方向的针对性检索，避免仅停留在传统关键词框架内。

总体而言，本次调研在检索策略设计与工具组合应用上取得了较为理想的效果，展现了我的文献调研能力，为后续深入研究人形机器人运动控制问题奠定了坚实的文献基础。